

NORFOLK, NE, USA

WMO: 725560

Lat: 41.986N

Lon: 97.435W

Elev: 1551

StdP: 13.89

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14941

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.6	-0.5	-13.1	2.9	-4.5	-8.5	3.7	0.4	33.9	18.1	30.8	19.1	9.5	320	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	93.8	74.2	90.8	73.4	88.0	72.1	77.8	88.4	76.2	86.6	74.4	84.6	14.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.7	137.7	84.5	72.9	129.5	82.9	71.2	121.8	81.2	42.5	88.7	40.7	86.6	39.1	84.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.9	24.9	22.1	DB	-13.7	98.2	5.2	3.8	-17.4	100.9	-20.4	103.2	-23.4	105.3	-27.1	108.0	(4)
(5)			WB	-14.3	80.6	5.0	1.3	-17.9	81.5	-20.8	82.3	-23.6	83.0	-27.2	83.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.9	22.9	26.5	38.4	49.5	60.7	70.8	75.4	72.8	65.2	51.7	37.9	25.7	(6)	
	DBStd	20.98	12.75	12.60	12.29	10.11	8.77	7.05	6.01	5.73	8.57	9.70	10.64	11.52	(7)	
	HDD50	3269	840	658	394	128	14	0	0	0	4	96	382	753	(8)	
	HDD65	6469	1305	1077	825	474	191	28	4	7	101	425	814	1218	(9)	
	CDD50	3240	0	1	34	112	345	623	789	708	461	148	19	0	(10)	
	CDD65	964	0	0	1	8	57	201	327	250	108	12	0	0	(11)	
	CDH74	10027	0	0	31	153	693	2077	3457	2306	1142	162	6	0	(12)	
	CDH80	3877	0	0	5	45	233	773	1499	872	401	49	0	0	(13)	
(14) Wind	WSAvg	10.4	10.7	11.1	11.7	12.4	10.9	9.8	8.7	8.4	9.6	10.4	10.5	10.3	(14)	
(15) Precipitation	PrecAvg	26.3	0.6	0.8	1.6	2.6	3.9	4.6	3.1	3.0	2.4	1.9	1.1	0.8	(15)	
	PrecMax	37.8	1.7	3.2	7.3	7.5	8.6	12.2	9.0	8.0	8.1	6.8	4.0	2.9	(16)	
	PrecMin	14.6	0.1	0.0	0.0	0.2	0.4	0.6	0.0	0.6	0.4	0.0	0.0	0.0	(17)	
	PrecStd	5.1	0.4	0.6	1.3	1.5	2.0	2.7	1.9	1.8	1.6	1.5	1.0	0.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.7	65.1	78.2	86.1	91.7	94.8	99.3	96.7	92.9	86.4	73.3	57.6	(19)	
		MCWB	44.9	49.1	56.2	61.4	66.9	73.7	74.9	74.6	71.4	65.2	55.0	45.9	(20)	
	2%	DB	50.3	55.3	70.8	78.2	86.0	90.8	94.7	91.7	88.7	78.0	66.1	51.5	(21)	
		MCWB	40.7	45.2	54.3	58.9	66.6	72.8	75.2	75.1	70.1	61.2	52.4	42.9	(22)	
	5%	DB	44.3	49.6	63.6	72.1	81.8	87.8	91.3	88.1	84.7	73.2	59.9	46.0	(23)	
		MCWB	37.3	41.5	51.9	56.2	64.4	71.3	75.1	73.6	68.8	58.6	49.0	38.8	(24)	
	10%	DB	39.5	44.2	57.7	67.2	77.1	84.6	88.1	85.0	80.9	68.3	54.3	41.1	(25)	
		MCWB	34.8	38.2	47.8	53.5	62.2	69.7	73.9	71.9	66.6	56.3	45.6	36.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.6	51.4	61.5	64.9	72.0	77.5	80.8	78.9	75.0	68.4	58.9	49.0	(27)	
		MCDB	57.5	60.5	70.6	80.4	83.3	88.8	91.4	89.8	86.0	79.7	66.5	55.1	(28)	
	2%	WB	41.2	46.0	57.1	61.1	69.1	75.4	78.6	77.0	72.8	64.4	53.6	42.7	(29)	
		MCDB	49.5	55.0	66.6	74.2	81.6	86.8	89.0	87.5	83.9	73.3	63.4	50.3	(30)	
	5%	WB	38.0	41.8	52.5	57.9	66.4	73.5	77.1	75.2	70.8	61.0	49.9	39.4	(31)	
		MCDB	43.6	49.1	63.4	70.2	78.1	84.6	87.4	85.2	81.4	69.9	59.4	45.0	(32)	
	10%	WB	35.0	38.3	48.0	54.9	63.9	71.6	75.2	73.6	68.8	57.7	45.9	36.2	(33)	
		MCDB	39.3	44.0	57.5	65.7	74.9	81.8	85.3	82.7	78.4	66.4	54.4	40.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	19.0	19.2	21.9	22.6	22.1	21.5	21.1	20.9	23.8	23.2	21.3	17.8	(35)	
		MCDBR	26.8	27.5	33.0	31.7	28.7	24.8	23.9	24.1	26.9	30.6	31.1	25.6	(36)	
	5% WB	MCWBR	18.4	18.1	18.7	16.3	13.1	11.2	10.6	11.1	12.3	15.7	18.2	17.6	(37)	
		MCWBR	25.5	26.2	29.6	28.2	24.4	21.5	20.5	21.4	22.8	24.5	28.3	24.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.275	0.288	0.313	0.378	0.397	0.411	0.414	0.415	0.370	0.323	0.285	0.274	(40)	
		taud	2.453	2.455	2.426	2.253	2.278	2.299	2.298	2.285	2.372	2.488	2.509	2.465	(41)	
	Ebn at Noon		280	294	297	282	279	274	272	268	273	277	275	271	(42)	
		Edh at Noon	24	28	32	42	42	41	41	40	34	27	22	22	(43)	
(44) All-Sky Solar Radiation	RadAvg		646	943	1240	1582	1849	2090	2106	1790	1450	1018	713	531	(44)	
	RadStd		52	67	101	101	130	145	108	129	98	114	52	51	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page